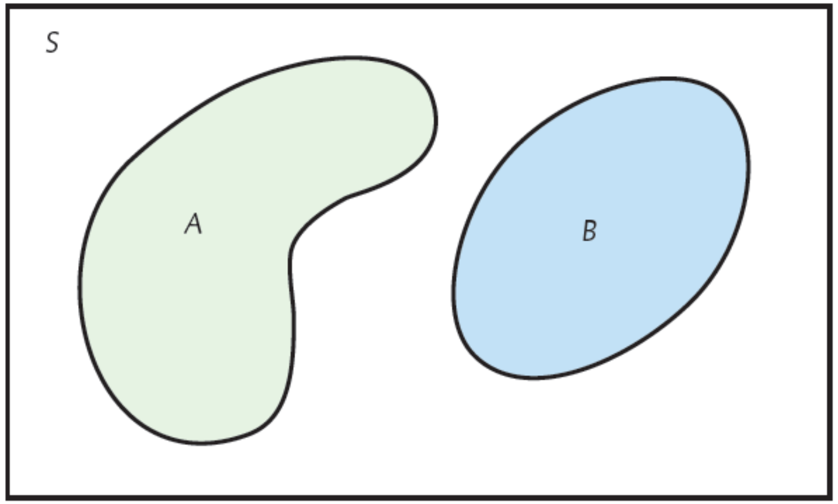
S. Mason Garrison

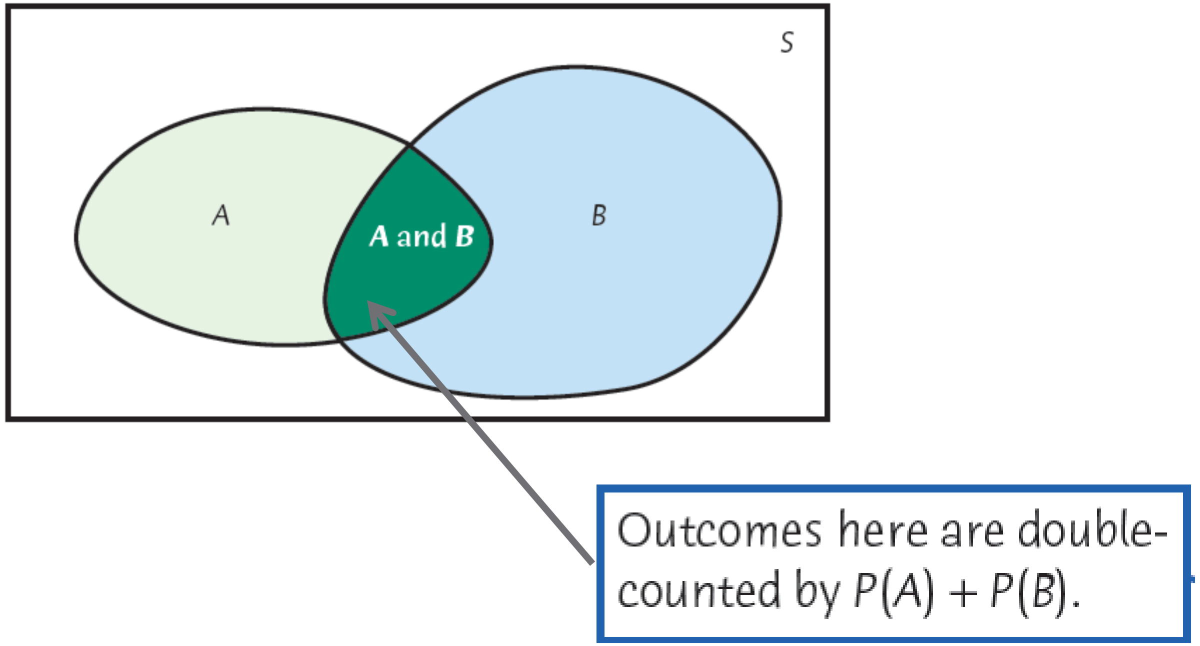
PROBABILITY, PT 2

May the odds be ever in your favor

THIS TIME… GENERAL RULES

The general addition rule





We know that if A and B are disjoint events:

P(A or B) = P(A) + P(B)

Addition rule for any two events

For any two events A and B:

P(A or B) = P(A) + P(B) – P(A and B)

The General Multiplication Rule

The probability that two events A and B occur together is

$$P(A \text{ and } B) = P(A)P(B \mid A)$$

where P(B | A) is the probability of B happening given that A has occurred.

This rule may be extended to any number of events:

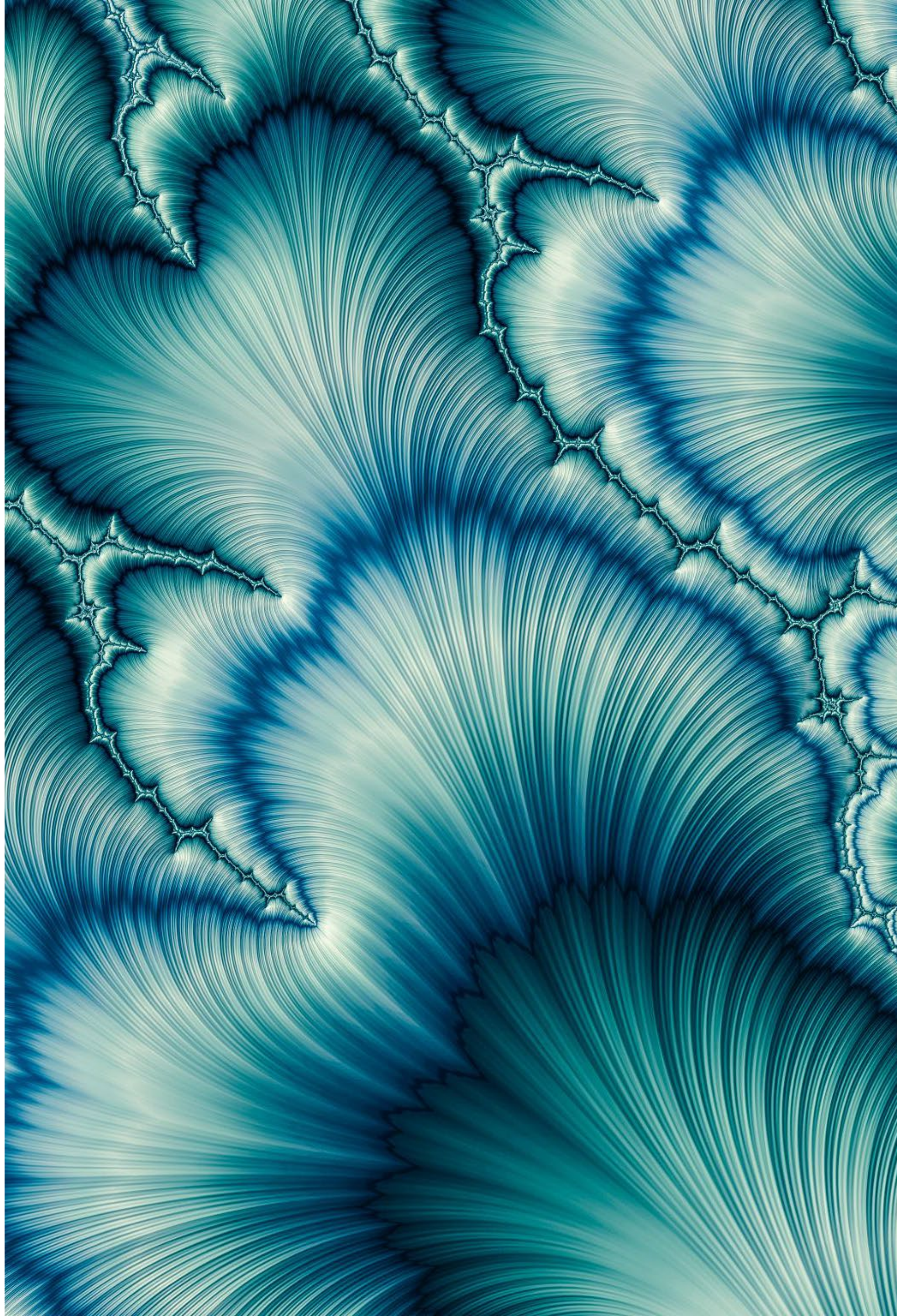
$$P(A \text{ and } B \text{ and } C) = P(A)P(B \mid A)P(C \mid \text{both } A \text{ and } B)$$

Conditional Probabilities Formalized

- Conditional probability is the probability of one event occurring given that another event has already occurred.
- It is represented as $P(A|B)$,
- which means the probability of A happening given that B has occurred.
- When P(A) > 0, the conditional probability of B given A is:

$$P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)}$$

Visual Example: http://setosa.io/conditional/



Conditional probability

[Visual Example: http://setosa.io/conditional/](http://setosa.io/conditional/)

A Visual explanation by Victor Powell for Setosa

A conditional probability is the probability of an event, given some other event has already occurred. In the below example, there are two possible events that can occur. A ball falling could either hit the red shelf (we'll call this event A) or hit the blue shelf (we'll call this event B) or both.

If we know the statistics of these events across the entire population and then were to be given a single ball and told "this ball hit the red shelf (event A), what's the probability it also hit the blue shelf (event B)?" we could answer this question by providing the conditional probability of B given that A occurred or P(B|A).

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Worked Example

Consider light truck and car sales in the U.S.:

	Domestic	Imported	Total
Light truck	712,700	187,000	899,700
Car	472,100	155,500	627,600
Total	1,184,800	342,500	1,527,300

If we consider only cars, what is the likelihood of a randomly selected sale being of a domestic vehicle?

$$P(\text{domestic}|\text{car}) =$$

In other words, we are finding the probability that a car sale is domestic, given that the vehicle is a car.

Setting Up the Problem

- We are given data about vehicle sales and need to find the probability that a randomly selected car is domestic.
- The formula for conditional probability is:
- $P(domestic \mid car) = \frac{P(domestic\ and\ car)}{P(car)}$
- Where:
  - P(domestic and car) is the probability of a sale being both domestic and a car.
  - P(car) is the probability of a sale being a car.

Step 1: Calculating Total Sales

- First, calculate the total number of sales:
  - Total sales=1,527,300, includes both light trucks and cars, domestic and imported.

	Domestic	Imported	Total
Light truck	712,700	187,000	899,700
Car	472,100	155,500	627,600
Total	1,184,800	342,500	1,527,300

Step 2: Calculating P(car)

- The number of car sales is 627,600.
  - Therefore, the probability of a randomly selected sale being a car is:
  - P(car) =627,600/1,527,300
  - =0.4108

	Domestic	Imported	Total
Light truck	712,700	187,000	899,700
Car	472,100	155,500	627,600
Total	1,184,800	342,500	1,527,300

Step 3: Calculating *P*(domestic and car)

- The number of domestic car sales is 472,100.
- So, the probability of a sale being both domestic and a car is:
- P(domestic and car)=
- 472,100/1,527,300=0.3091

	Domestic	**Importe d**	**Total**
Light truck	712,700	187,000	899,700
Car	472,100	155,500	627,600
Total	1,184,800	342,500	1,527,300

Step 4: Conditional Probability

- Now, we can calculate the conditional probability:

- $P(\text{domestic}|\text{car}) = \dfrac{472{,}100}{627{,}600} = \dfrac{\frac{472{,}100}{1{,}527{,}300}}{\frac{627{,}600}{1{,}527{,}300}} = \dfrac{P(\text{domestic and car})}{P(\text{car})}$

- This means that if a randomly selected vehicle is a car, there is a 75.25% chance it is domestic.

Now that we know the basics

- Let's apply that knowledge to those two Renaissance Dice Games
- Reminder:
  - Game 1
    - Bet whether within 4 rolls, the die will come up a 6
  - Game 2
    - Bet whether within 24 rolls, the dice will roll double 6s

Game 1

- Game: Bet whether within 4 rolls, the die will come up a 6
- p(1/6 | 4 rolls)
  - ~~1/6*1/6*1/6*1/6~~
    - (would be the case if we rolled a 6 each time)
  - ~~5/6*5/6*5/6*5/6~~
    - (would be the case if we failed to roll a 6 each time)
  - 1-5/6*5/6*5/6*5/6
    - (would be the case if we didn't fail to roll a 6)
  - = 1-(5/6)^4
  - = 0.5177469
- To be profitable, the house should bet that 6s will come up
  - In 100 games, they'd expect to win 51.8 of them.

Game 2

- Game: Bet whether within 24 rolls, the dice will roll double 6s
- p(1/36|24 rolls)
  - 1-35/36*35/36*35/36*35/36……..
    - (would be the case if we didn't fail to roll two 6s)
  - = 1-(35/36)^24
  - = 0.4914039
- To be profitable, the house should bet that double 6s won't come up
  - Otherwise, in 100 games, they'd expect to only win 49.1 of them.
  - Instead, they'll win 100-49.1 of them

Wrapping Up…

Practical Implications

Expected Values

- Expected value is exactly what you might think it means intuitively:
  - the **return you can expect for some kind of action**

- For the two dice games, I showed the expected value for 100 games.

- The basic expected value formula is the
  - probability of an event
  - multiplied by the number of times the event happens
  - **(P(x) * n)**.

Expected Values

- For example, if you take a 20-question multiple-choice test with A,B,C,D as the answers, and you guess all “A”,
  - then you can expect to get 25% right (5 out of 20).
- The **math** behind this kind of expected value is:
- The probability (P) of getting a question right if you guess: .25
- The number of questions on the test (n)*: 20
  - **P x n = .25 x 20 = 5**

Expected Value for Multiple Events

- Calculating expected value (EV) gets more complicated in real life.
- For example, You buy two $10 raffle tickets for a new car valued at $15,000.
- Two thousand tickets are sold. What is the EV of your gain?
- The formula for calculating the EV where there are **multiple probabilities** is:
  - $E(X) = \sum(X * P(X))$
- The expected value for your 2 $10 raffle tickets are:
  - $15,000*1/2000+ $15,000*1/2000 =$15
- Would you buy the tickets?
  - At what ticket price would you be indifferent?

Personal probability

- A personal probability of an outcome is a number between 0 and 1 that expresses an individual's judgment of how likely the outcome is.

- To be legitimate, a personal probability must obey the rules of probability.
  - $0 \leq P(A) \leq 1$
  - $P(S) = 1$
  - $P(A \text{ or } B) = P(A) + P(B)$ IF disjoint
  - $P(\text{not } A) = 1 - P(A)$

- It may not match another individual's personal probability of an event.

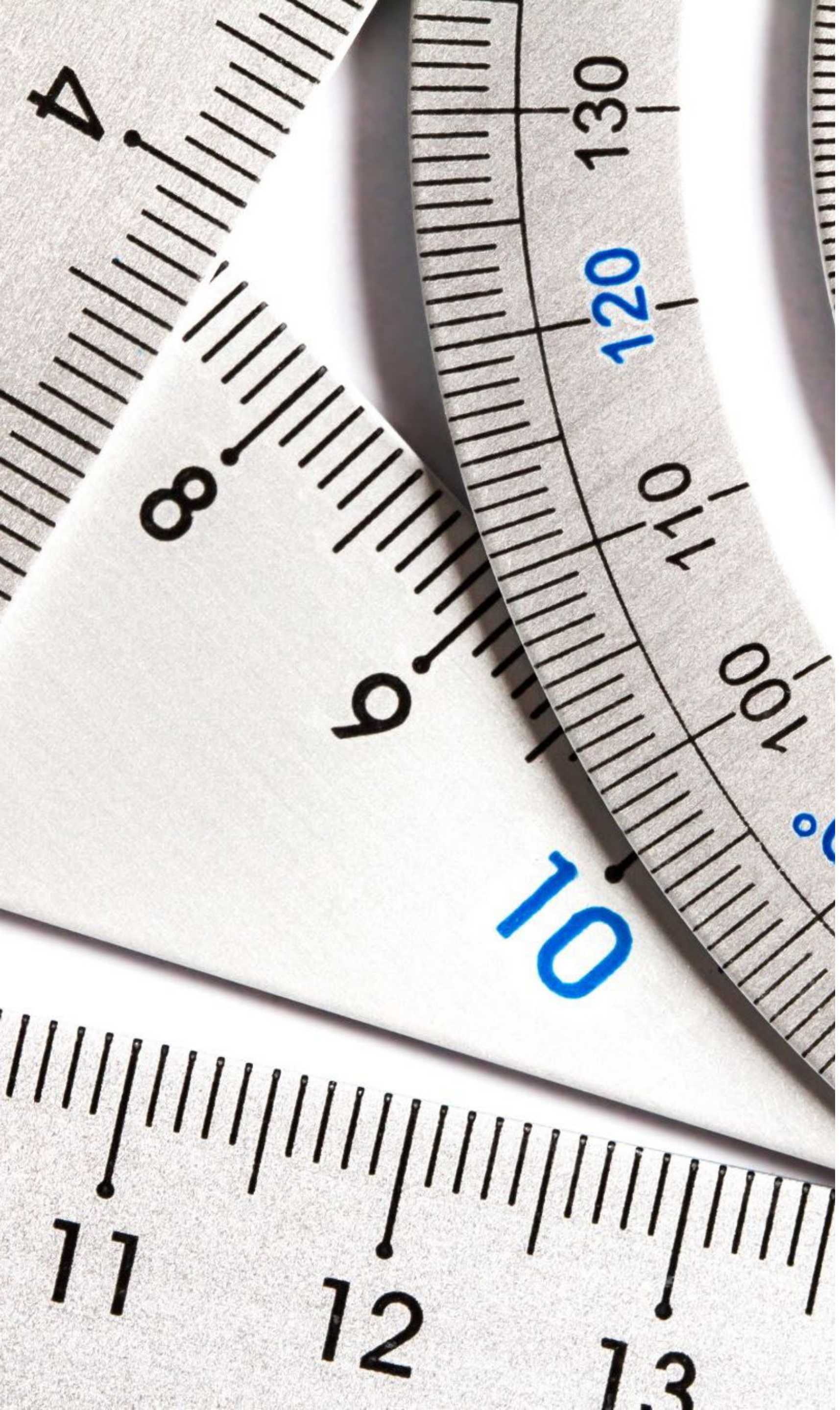
Example: The Linda Problem

- Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.
- Which is more probable?
  - Linda is a bank teller.
  - Linda is a bank teller and is active in the feminist movement.

The Conjunction Fallacy

- The Linda Problem is an illustration of the conjunction fallacy.
- The majority of those asked chose option 2. (Tversky and Kahneman, 1982)
  - However, the probability of two events occurring together (in "conjunction") is always less than or equal to the probability of either one occurring alone
  - formally, for two events A and B this inequality could be written as
    - $P(A \text{ and } B) \leq P(A)$ and $P(A \text{ and } B) \leq P(B)$
  - Let's chose a really low probability that Linda is a bank teller $P(BT)=.05$,
    - and a high probability that she's a feminist $P(F)=.95$
  - We use the multiplication rule (and assume independence).
    - P (bank teller *and* feminist) = 0.05×0.95 or 0.0475

Wrapping Up…



Bayes Rule

Bayes's rule

- Bayes's rule provides a unifying structure and is a basic result in an important area of statistics known as Bayesian statistics.
- Example: In the last example, we were given P(online dating | 18 to 24) = 0.27 (if we chose a random 18- to 24-year-old, the probability that they had used an online dating site was 0.27). But what if we want P(18 to 24 | online dating)?

Bayes's rule

- Suppose $B_1, B_2, \ldots, B_n$ are disjoint events with positive probabilities, and the sum of these probabilities is 1.
- If A is any event with probability greater than 0, then

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_n)P(B_n)}$$

Bayes's rule (example, part I)

Looking only at adult Internet users, aged 18 and over, about 15% are 18–29 years old, another 26% are 30–49 years old, and the remaining 59% are 50 and over. The Pew Internet and American Life Project finds that 68% of Internet users aged 18–29 have posted photos or videos they have found online, along with 54% of those aged 30–49 and 26% of those 50 or older.

What percent of adult Internet users who post photos or videos they have found online are aged 18–29? Are aged 30–49? Are 50 or older?

If $B_1 = \{\text{aged } 18\text{–}29\}$ then $P(B_1) = 0.15$
If $B_2 = \{\text{aged } 30\text{–}49\}$ then $P(B_2) = 0.26$
If $B_3 = \{\text{aged 50 and older}\}$ then $P(B_3) = 0.59$
If $A = \{\text{post photos or videos}\}$ then $P(A \mid B_1) = 0.68$
and $P(A \mid B_2) = 0.54$
and $P(A \mid B_3) = 0.26$

We want to find $P(B_1 \mid A)$, $P(B_2 \mid A)$, and $P(B_3 \mid A)$.

Bayes's rule (example, part II)

(What percent of adult Internet users who post photos or videos they have found online are aged 18–29? Are aged 30–49? Are 50 or older?)

Using Bayes's rule:

$$P(B_1|A) = \frac{P(A|B_1)P(B_1)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3)}$$

$$= \frac{(0.68)(0.15)}{(0.68)(0.15) + (0.54)(0.26) + (0.26)(0.59)}$$

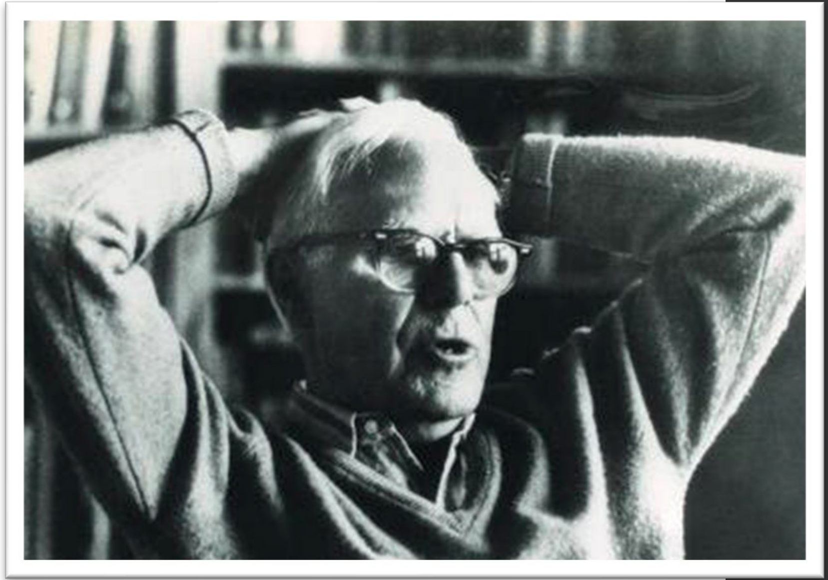
$$= 0.2577$$

Similar calculations show $P(B_2 \mid A) = 0.3547$ and $P(B_3 \mid A) = 0.3876$.

About 26% of adults who post photos or videos that they have found online are between 18 and 29 years old, 35% are between 30 and 49 years old, and the remaining 39% are aged 50 or over.

Boy or Girl Paradox (Gardener, 1959)

- Martin Gardner published one of the earliest variants of this paradox in Scientific American. He phrased the paradox as follows:
  - "Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?
  - Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?"
- Gardner gives the answers as ½ and 1/3.
  - But, there's much debate as to whether the second question's
    - answer is ½ or 1/3



1. What is the probability that both children are girls?

Older Child	Younger Child
Girl	Girl
Girl	Boy
Boy	Girl
Boy	Boy

- “Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?”
- Break down the possible options in the sample space

Older Child	Younger Child
Girl	Girl
Girl	Boy
~~Boy~~	~~Girl~~
~~Boy~~	~~Boy~~

1. What is the probability that both children are girls?

- "Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?"
- Eliminate options

1. What is the probability that both children are girls?

- “Mr. Jones has two children. The older child is a girl. What is the probability that both children are girls?”
- Eliminate options

Older Child	Younger Child
Girl	Girl
Girl	Boy
Boy	Girl
Boy	Boy

2. What is the probability that both children are boys?

- Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?
- If we formulate the question as:
  - From all families with two children,
    - one child is selected at random,
      - and the sex of that child is specified to be a boy.

2. What is the probability that both children are boys?

- Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?

- If we formulate the question as:
  - From all families with two children,
    - one child is selected at random,
      - and the sex of that child is specified to be a boy.

2. What is the probability that both children are boys?

- Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?
- If we formulate the question as:
  - From all families with two children,
    - at least one of whom is a boy,
    - a family is chosen at random.

2. What is the probability that both children are boys?

- Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?
- If we formulate the question as:
  - From all families with two children,
    - at least one of whom is a boy,
    - a family is chosen at random.

2. What is the probability that both children are boys?

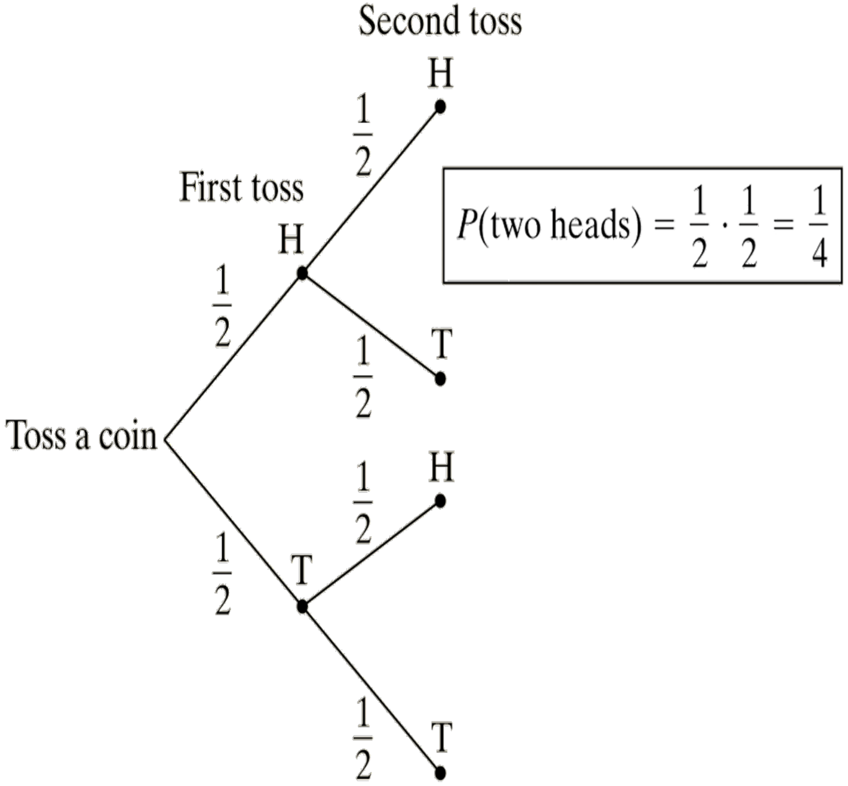
- Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?
- If we formulate the question as:
  - From all families with two children,
    - at least one of whom is a boy,
    - a family is chosen at random.

Alternative Approaches

Tree diagrams

- We learned how to describe the sample space S of a chance process
- Another way to model chance behavior that involves a sequence of outcomes is to construct a tree diagram.

Consider flipping a coin twice. What is the probability of getting two heads?



Sample Space

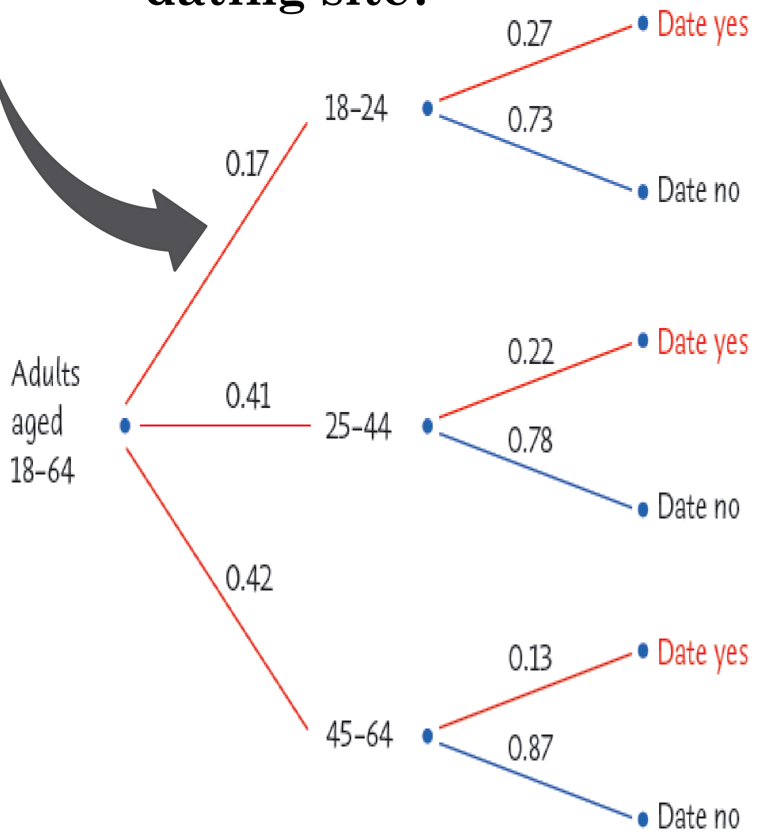
HH HT TH TT

So, P(two heads) = P(HH) = 1/4.

Tree diagrams (example)

Looking only at adults under 65, about 17% are 18–24 years old, another 41% are 25–44 years old, and the remaining 42% are 45–64 years old. Pew Research reports that 27% of those aged 18–24 have used online dating sites, along with 22% of those aged 25–44 and 13% of those aged 45–64.

What percent of American adults under 65 have used an online dating site?



$\rightarrow P(18 \text{ to } 24) = 0.17,\ P(\text{online dating} | 18 \text{ to } 24) = 0.27$

$$\begin{aligned} &P(18 \text{ to } 24 \text{ and online dating}) \\ &\quad = P(18 \text{ to } 24) P(\text{online dating} \mid 18 \text{ to } 24) \\ &\quad = (0.17)(0.27) \\ &\quad = 0.0459 \end{aligned}$$

$$\begin{aligned} &P(\text{online dating}) \\ &\quad = P\begin{pmatrix} 18 \text{ to } 24 \text{ and online dating or} \\ 25 \text{ to } 44 \text{ and online dating or} \\ 45 \text{ to } 64 \text{ and online dating} \end{pmatrix} \\ &\quad = (0.27)(0.17) + (0.41)(0.22) + (0.42)(0.13) \\ &\quad = 0.1907 \end{aligned}$$

Simulations

- Blackjack
- Rock Paper Scissors
- Texas Hold em
- Lottery (Adam Lamers)
- Lottery (Ian Neath)

Wrapping up…